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Trajectory Constraint Solution Algorithm

15 July 2014

Introduction

In some sounding rocket competitions, notably the annual Intercollegiate Rocket Engineering Competition (IREC) sponsored by the Experimental Sounding Rocket Association (ESRA) the contestants are asked to achieve apogee altitude and landing range as close as possible to specified aim points. Usually the rocket itself is fully defined a priori. That is, the thrust curve, the weight statement, and the drag coefficient are fully known and defined at the start of the process. The purpose of this memo is to document an iterative, man-in-the-loop algorithm for solving this problem. A man-in-the-loop algorithm is needed because many trajectories are significantly nonlinear. The control variables available are the drogue parachute deployment time and the initial launcher quadrant elevation angle (QE).

Several problems can arise in this process. In no particular order these are:

1. The sensitivity determinant is singular, or nearly so. In this case the corrections to the initial estimates of the drogue deployment time and launcher QE become very large. The remedy is to create a new starting point and try again.
2. Suppose the solution is a QE too high from a range safety perspective. Then select the highest QE allowed and find the drogue deployment time that yields the desired apogee.
3. What if there is no solution? That is, the rocket lacks the energy to satisfy both constraints. In this case, satisfy the apogee constraint while maximizing the impact range.

These issues will be addressed in that order in the remainder of this memo.

Notation

Mnemonic	Definition
TD	Drogue parachute deployment time, sec after liftoff
QE	Initial launcher quadrant Elevation angle above the horizon
QE_{MAX}	The maximum QE allowable for range safety reasons
h^*	Apogee altitude, feet
r^{**}	Landing range from the launcher, feet
h_T	$=\partial h^* / \partial TD$
h_Q	$=\partial h^* / \partial QE$
r_T	$=\partial r^{**} / \partial TD$
r_Q	$=\partial r^{**} / \partial QE$
$\bar{h}$	Desired apogee altitude, feet
Mnemonic	Definition
$\bar{r}$	Desired landing range, feet

λ Increment dilution factor

Nominal Analysis

Begin by guessing the nominal values of drogue deployment time and quadrant elevation angle. Make a trajectory simulation using these values. Then make two more simulations, one with slightly different drogue deployment time and the other with slightly different quadrant elevation angle. Increments of 1 second and 1 degree are appropriate. Label the values for the first trajectory with the subscript 1, and the next two with subscripts 2 and 3 respectively. Then, estimate the partial derivatives:

$$r_Q = \frac{r_3^{**} - r_1^{**}}{QE_3 - QE_1},$$

$$h_Q = \frac{h_3^{*} - h_1^{*}}{QE_3 - QE_1},$$

$$r_T = \frac{r_2^{**} - r_1^{**}}{TD_2 - TD_1}, \text{ and}$$

$$h_T = \frac{h_2^{*} - h_1^{*}}{TD_2 - TD_1}.$$

Clearly, trajectory 2 has the same quadrant elevation angle as trajectory 1. Similarly, trajectory 3 has the same drogue deployment time as trajectory 1.

The next trajectory, labeled with subscript 4, is the first with adjusted estimates on drogue deployment time and launcher quadrant elevation angle. If these adjustments are small, a linear Taylor series will provide the tool needed:

$$\bar{h} = h_1^{*} + h_Q \Delta QE + h_T \Delta TD, \text{ and}$$

$$\bar{r} = r_1^{**} + r_Q \Delta QE + r_T \Delta TD, \text{ where}$$

$$\Delta QE = QE_4 - QE_1, \text{ and}$$

$$\Delta TD = TD_4 - TD_1.$$

Here $\bar{h}$ and $\bar{r}$ are the desired apogee and landing range.

These can be solved to give:

$$\Delta QE = \lambda \frac{(\bar{h} - h_1^*) r_T - (\bar{r} - r_1^{**}) h_T}{S}, \text{ and}$$

$$\Delta TD = \lambda \frac{(\bar{r} - r_1^{**}) h_Q - (\bar{h} - h_1^*) r_Q}{S}, \text{ where}$$

$$S = h_Q r_T - h_T r_Q$$

As already noted, the sensitivity determinant S should always be checked to ensure it's not too small. Here λ is a factor (≈ 0.5) intended to slow down the rush to a solution to prevent overshoot.

From here on it's straightforward. Run a new trajectory simulation with QE_4 and TD_4. A new set of adjustments can be found using the above process. Set the trajectory 4 just found as the new trajectory 1, and iterate.

Excessive Quadrant Elevation Angle

Consider what can happen if the QE resulting from the above process were too high. This implies the desired landing range is too close to the launcher, and, will, therefore, be unacceptable to range safety. In such a case, select the maximum allowable QE, and iterate for the drogue deployment time:

$$QE = QE_{MAX}, \text{ and}$$

$$\Delta TD = \frac{\bar{h} - h_1}{h_T}.$$

Insufficient Energy

Another potential problem is that no viable solution meeting both constraints can be found. In essence, the rocket does not have enough energy to reach both the desired altitude and landing range. Then, because the apogee constraint is the more important one in the ESRA culture, continue to try to satisfy it while seeking the greatest possible range. To find the maximum range possible while still reaching the constraint altitude, simply set the coast time to a very large value (e.g., 999 sec) and iterate to find the QE that maximizes landing range while achieving the desired apogee.

These results have been coded in an Excel® spreadsheet named TRAJCON.xls. The example below is based on a desired apogee of 10000 feet AGL, a desired landing range of 5000 feet and $\lambda = 0.5$. We start with an initial iteration based on experience/guesswork. The following table summarizes the initial steps in the trajectory design based on SKYAERO runs:

Iteration	Apogee Altitude, ft	Landing Range, ft	Quadrant Elevation, deg	Drogue Deployment Time, sec
1	11237.65	2822.23	85.0	13.5

2	10473.11	3689.75	82.216	10.163
3	10243.85	3968.63	81.282	9.530
4	10090.37	4454.73	80.013	9.386
5	10047.32	4729.53	79.371	9.475
6	10023.51	4865.98	79.049	9.517

Note that the drogue deployment time is measured from booster burnout. Each iteration requires 3 SKYAERO runs, one for the nominal case and the other two for the partial derivatives.

Finally, for this case the QE that maximizes range while still meeting the 10000 foot apogee constraint is about73.34°.

Resolution

When the iteration is close to a solution, stop and recall that drogue deployment is altitude based using an altimeter. Commercial digital altimeters typically have a granularity or resolution of 100 feet. From here on fix the drogue deployment time to that corresponding to the nearest 100 feet, and iterate to find the quadrant elevation angle matching the desired apogee. It may be necessary to do this with several drogue deployment altitudes to find the one matching the desired landing range most closely. In the above example, the best drogue deployment altitude is 9300 feet.